

Unit 2: Elements Simple Random Sampling

- A. Implementation
- B. Precision & inference
- C. Sample size

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A. Implementation

- Population of N , sample of n : $\binom{N}{n} = \frac{N!}{n!(N-n)!}$
- Create all samples of size n from N , select one at random
- Alternatively, random selection using mechanical device
 - Table of random numbers, using labels i to identify selected units

Seq. No.	Division	Sex	Ran	Salary
1	Eng&Prof	m	3	\$88
2	Medicine	f	3	\$45
3	Medicine	m	3	\$57
4	Medicine	m	1	\$133
5	Eng&Prof	f	2	\$71
6	Lit&SocSci	m	1	\$113
7	Medicine	f	3	\$65
8	Biol&Sci	m	3	\$47
9	Medicine	f	3	\$39
10	Lit&SocSci	m	1	\$74
11	Medicine	m	1	\$88
12	Lit&SocSci	m	1	\$62
13	Lit&SocSci	m	1	\$49
14	Medicine	m	3	\$88
15	Medicine	m	1	\$181
16	Eng&Prof	m	3	\$63
17	Medicine	m	2	\$94
18	Eng&Prof	m	1	\$91
19	Medicine	m	1	\$60
20	Eng&Prof	m	3	\$55
21	Biol&Sci	m	2	\$55
22	Medicine	f	1	\$106
23	Medicine	m	1	\$116
24	Medicine	m	3	\$79
25	Lit&SocSci	m	1	\$61
26	Lit&SocSci	f	3	\$37
27	Medicine	m	2	\$72
28	Eng&Prof	m	1	\$105
29	Medicine	m	2	\$79
30	Medicine	m	1	\$61
31	Medicine	m	1	\$86
32	Biol&Sci	m	1	\$103
33	Lit&SocSci	m	1	\$48
34	Eng&Prof	m	2	\$64
35	Eng&Prof	m	1	\$78
36	Medicine	f	2	\$53
37	Biol&Sci	m	1	\$85
38	Eng&Prof	m	1	\$61
39	Medicine	m	1	\$106
40	Lit&SocSci	m	2	\$60
41	Biol&Sci	f	1	\$73
42	Medicine	m	1	\$70
43	Medicine	f	3	\$32
44	Lit&SocSci	m	2	\$49
45	Eng&Prof	m	3	\$43
46	Medicine	m	1	\$75
47	Lit&SocSci	m	1	\$92
48	Medicine	m	2	\$107
49	Biol&Sci	m	2	\$57
50	Medicine	m	2	\$114
51	Eng&Prof	m	3	\$55
52	Biol&Sci	m	1	\$49

53	Eng&Prof	m	3	\$57
54	Medicine	m	1	\$118
55	Medicine	m	3	\$84
56	Eng&Prof	m	3	\$52
57	Medicine	m	3	\$64
58	Eng&Prof	m	1	\$75
59	Medicine	f	1	\$87
60	Eng&Prof	m	3	\$58
61	Medicine	f	3	\$39
62	Medicine	m	3	\$69
63	Medicine	f	2	\$46
64	Eng&Prof	f	1	\$86
65	Medicine	m	3	\$87
66	Medicine	m	3	\$59
67	Eng&Prof	f	3	\$44
68	Medicine	m	2	\$123
69	Lit&SocSci	f	3	\$37
70	Lit&SocSci	m	1	\$106
71	Lit&SocSci	m	1	\$91
72	Lit&SocSci	m	1	\$78
73	Biol&Sci	m	1	\$77
74	Medicine	m	1	\$90
75	Eng&Prof	m	2	\$71
76	Medicine	f	3	\$42
77	Medicine	f	2	\$59
78	Eng&Prof	m	2	\$49
79	Biol&Sci	m	1	\$83
80	Lit&SocSci	m	1	\$34
81	Medicine	f	3	\$42
82	Medicine	m	2	\$97
83	Medicine	m	1	\$109
84	Lit&SocSci	f	2	\$48
85	Medicine	m	1	\$47
86	Eng&Prof	m	2	\$45
87	Medicine	m	3	\$83
88	Medicine	m	2	\$51
89	Biol&Sci	m	1	\$78
90	Lit&SocSci	m	1	\$70
91	Eng&Prof	f	2	\$46
92	Eng&Prof	m	1	\$85
93	Lit&SocSci	m	1	\$53
94	Medicine	f	3	\$40
95	Eng&Prof	m	1	\$87
96	Lit&SocSci	m	1	\$71
97	Medicine	m	1	\$75
98	Biol&Sci	m	1	\$85
99	Lit&SocSci	m	2	\$50
100	Medicine	m	3	\$118
101	Lit&SocSci	m	2	\$55
102	Medicine	m	3	\$80
103	Eng&Prof	m	1	\$114
104	Lit&SocSci	m	1	\$63
105	Medicine	m	1	\$112

Table of Random Numbers

	1-5	6-10	11-15	16-20	21-25	26-30	31-35	36-40	41-45	46-50
1	57982	03250	09488	12085	38325	01091	03826	77648	57881	04677
2	34104	88576	32549	09603	73860	43758	05470	38344	04492	10730
3	01943	29119	72315	70334	98029	80936	37792	45017	18819	84787
4	25329	02138	52670	11692	03654	91109	17357	27013	32767	38642
5	23869	85541	29993	57693	84597	43545	23014	06089	52569	93160
6	69419	03843	33691	19541	80307	96350	62760	07467	12970	83870
7	14078	76539	08538	17888	65481	03995	63362	29308	23196	51186
8	14339	07381	13958	46562	75860	66994	97734	83135	77941	39411
9	97670	49015	97498	01260	63189	29044	11089	81803	87889	81923
10	15279	11006	82780	74184	34105	09246	31268	91081	99324	68381
11	97947	01993	12603	63485	16937	06190	77536	18598	92322	21257
12	00631	55073	73183	05961	16402	39801	96560	88053	70254	73836
13	30246	12514	24792	46736	61024	87174	63045	35815	57531	41129
14	74016	75832	78113	18374	78851	46324	78629	54165	73712	08415
15	53669	78559	66231	29085	78506	80934	88520	86912	17774	29711
16	98280	13316	23734	63720	50355	74201	17325	69300	28342	91793
17	32396	30527	10030	69065	87877	81658	23343	87272	74683	86065
18	05708	99540	11289	09030	94001	05587	41643	40782	25630	48366
19	04004	49911	47941	81165	14505	07432	79939	01309	79165	57871
20	84424	61863	81655	87420	76149	74712	81555	82645	05850	54779
21	84546	16063	09051	31787	57235	27302	83322	48361	11630	80174
22	76882	00300	45868	17098	53385	83121	70986	70598	91543	44137
23	98169	25789	99043	73890	35327	91723	77932	77971	41355	19562
24	33621	78638	27009	29610	20171	74951	34482	89391	81210	86209
25	61507	79038	38331	70320	73347	57565	19920	41574	68935	79233
26	20920	81284	01289	66936	60782	71442	63257	45294	11141	15333
27	78861	84019	31418	64943	42169	88995	48423	95960	27581	11679
28	08307	99947	51173	87234	60693	06143	13459	59514	79507	38474
29	99441	02441	66284	05523	19477	51695	58978	92845	69511	81076
30	79407	56191	54855	83823	33322	96478	39901	82641	38556	56564
31	34643	05408	67404	45254	37951	12189	76651	51762	59986	83559
32	79071	00901	91791	64053	47239	64456	10573	91720	32152	70961
33	39776	75808	19351	39680	75676	29758	11181	48497	06845	08877
34	39404	45516	97619	49788	96815	22561	01201	50766	19787	53701
35	48094	09146	14671	56419	47127	53938	80707	35195	31953	27177
36	18410	40494	13931	61713	27618	34620	72264	38473	21435	08084
37	67160	05126	99743	18590	94419	32835	22290	76000	81239	43728
38	08555	37978	24194	23612	83749	24952	05356	66661	57819	99209
39	68035	97917	09711	82240	05453	43354	44223	19470	20455	93951
40	12989	19518	39150	20121	60930	48430	29572	66745	10380	89898
41	54634	90880	59787	00767	25687	88036	32752	27228	52429	14533
42	22603	31768	32309	68558	90881	87299	30044	06782	43324	65189
43	59743	43696	26720	27716	39721	74953	90138	90812	86546	07429
44	97687	86485	58473	00533	12837	74736	31091	42786	44844	10143
45	45106	28013	06476	46786	61360	15114	00315	45583	85248	59929
46	05655	11433	54983	86786	26287	75827	57443	70597	38351	91876
47	51610	48745	02202	10037	94816	87585	15188	64636	70648	50783
48	86277	69022	37506	91519	22285	04022	49127	77337	93702	80773
49	51243	81450	22865	61165	40356	58407	82449	40460	28950	00041
50	72699	19280	12900	57824	58612	00859	35035	82783	15422	45453

A. Implementation

- Population of N , sample of n : $\binom{N}{n} = \frac{N!}{n!(N-n)!}$
- Create all samples of size n from N , select one at random
- Alternatively, random selection using mechanical device
 - Table of random numbers, using labels i to identify selected units
 - Software using pseudo-random numbers
 - Not haphazard
- Base procedure
 - SRS – without replacement
 - SRS seldom used in practice, even from a simple frame
 - All samples of n distinct elements equally likely

Probabilities of selection

- Draw by draw ...
- With replacement (Unrestricted Random, URS):
 - $p_{ij} = 1/N$, for draw j and element $i = 1, \dots, N$
- For SRS, across all draws, $\pi_i = f = n/N$:

$$p_1 = \frac{1}{N}$$

$$p_2 = (1 - p_1) \left[\frac{1}{N-1} \right] = \left[\frac{N-1}{N} \right] \left[\frac{1}{N-1} \right] = \frac{1}{N}$$

$$p_3 = (1 - p_1)(1 - p_2) \left[\frac{1}{N-2} \right] = \frac{1}{N}$$

Sample estimates

$$y_1, y_2, \dots, y_n \quad y_1 + y_2 + \dots + y_n = \sum_{i=1}^n y_i = y$$

$$\bar{y} = \sum_{i=1}^n y_i / n = y/n$$

$$s^2 = \left[(y_1 - \bar{y})^2 + \dots + (y_n - \bar{y})^2 \right] / (n-1)$$

$$= \sum_{i=1}^n (y_i - \bar{y})^2 / (n-1)$$

$$= \left(\sum_{i=1}^n y_i^2 - \frac{y^2}{n} \right) / (n-1)$$

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B. Precision & inference

$$\bar{y} = y/n$$

$$Var(\bar{y}) = \frac{1}{\binom{N}{n}} \sum_{s=1}^{\binom{N}{n}} (\bar{y}_s - \bar{Y})^2 = \frac{(1-f)}{n} S^2$$

$$\begin{aligned} \text{var}(\bar{y}) &= \frac{(1-f)}{n} s^2 = \\ &= \frac{(1-f)}{n} \frac{\sum_{i=1}^n (y_i - \bar{y})^2}{(n-1)} = \frac{\left(1 - \frac{n}{N}\right)}{n} \frac{\left(\sum_{i=1}^n y_i^2 - \frac{y^2}{n}\right)}{(n-1)} \end{aligned}$$

$$se(\bar{y}) = \sqrt{\text{var}(\bar{y})}$$

Other estimates: Population Total

$$\hat{Y} = Fy = \left(\frac{N}{n}\right)y = N\bar{y}$$

$$Var(\hat{Y}) = F^2 Var(y) = N^2 Var(\bar{y})$$

$$var(\hat{Y}) = N^2 var(\bar{y}) = N^2 \frac{(1-f)}{n} s^2$$

$$se(\hat{Y}) = N \sqrt{var(\bar{y})}$$

Other estimates: Proportions

- Values of Y are zero or 1:

$$p = \bar{y}$$

$$Var(p) = \frac{(1-f)}{n} S^2 = \frac{(1-f)}{n} \left(\frac{N}{N-1} \right) P(1-P)$$

$$var(p) = \frac{(1-f)}{n} s^2 = \frac{(1-f)}{n} \left(\frac{n}{n-1} \right) p(1-p) = \frac{(1-f)}{n-1} p(1-p)$$

Inference

- Standard deviation & element variance

$$s = \sqrt{s^2}$$

- Standard error & sampling variance

$$se(\bar{y}) = \sqrt{\text{var}(\bar{y})} = \sqrt{\frac{(1-f)}{n} s^2} = s \sqrt{\frac{(1-f)}{n}}$$

- Confidence interval

$$\bar{y} \pm t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} = \bar{y} \pm t_{1-\alpha/2, n-1} \sqrt{\frac{(1-f)}{n} s^2}$$

$$p \pm t_{1-\alpha/2, n-1} \sqrt{\text{var}(p)} = p \pm t_{1-\alpha/2, n-1} \sqrt{\frac{(1-f)}{n-1} p(1-p)}$$

Without Replacement Sampling

- Population of $N = 5$, **without** replacement selection of $n = 3$:

(1) Y_1, Y_2, Y_3 $\bar{y}_1$

(2) Y_1, Y_2, Y_4 $\bar{y}_2$

•

•

•

(10) Y_3, Y_4, Y_5 $\bar{y}_{10}$

$$Var(\bar{y}) = \frac{1}{10} \sum_{s=1}^{10} \left(\bar{y}_{(s)} - \bar{Y} \right)^2 = \frac{(1-f)}{n} S^2$$

With Replacement Sampling

- Population of $N = 5$, **with** replacement selection of $n = 3$:

(1) Y_1, Y_1, Y_1 $\bar{y}_1$

(2) Y_1, Y_1, Y_2 $\bar{y}_2$

•

•

•

(125) Y_5, Y_5, Y_5 $\bar{y}_{125}$

$$Var(\bar{y}) = \frac{1}{125} \sum_{s=1}^{125} \left(\bar{y}_{(s)} - \bar{Y} \right)^2 = \frac{1}{n} S^2$$

Without or With?

- Population of $N = 5$, **without** and **with** replacement selection of $n = 3$:

(1) Y_1, Y_2, Y_3

(2) Y_1, Y_2, Y_4

•

•

•

(10) Y_3, Y_4, Y_5

$$Var(\bar{y}) = \frac{(1-f)}{n} S^2$$

(1) Y_1, Y_1, Y_1

(2) Y_1, Y_1, Y_2

•

•

•

(125) Y_5, Y_5, Y_5

$$Var_{URS}(\bar{y}) = \frac{1}{n} S^2$$

Proportions: useful properties

$$s^2 = \left(\frac{n}{n-1} \right) p(1-p) \approx p(1-p)$$

$$\text{var}(p) = \frac{(1-f)}{(n-1)} p(1-p)$$

P	$S^2 \approx P \times Q$	$S \approx \sqrt{P \times Q}$
0.50	0.250	0.500
0.60	0.240	0.490
0.70	0.210	0.458
0.95	0.048	0.218

Proportions: useful properties

- What does this mean for us?
- Projection: estimating standard errors before we draw a sample
 - For proportions, p can project s without large error

p	$s^2 \approx p \times q$	$s \approx \sqrt{p \times q}$
0.50	0.250	0.500
0.60	0.240	0.490
0.70	0.210	0.458
0.95	0.048	0.218

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C. Sample size

- Sampling error depends very little on N
 - Unless f greater than 0.05
 - (Approximately) Same precision for $n = 1,000$ in Ann Arbor or the People's Republic of China
- Sample size determination in SRS
 - Specify desired level of precision
 - Obtain value for element variance
 - Solve for n

Computation

- Desired precision: $\hat{V}^2 = \left(1 - \frac{n}{N}\right) \frac{S^2}{n}$
- Solve algebraically for n : $n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$
- Ignoring the *fpc*: $n = \frac{\hat{S}^2}{\hat{V}^2}$

Desired precision: confidence interval or margin of error (e)

- The desired level of precision can also be specified by the width of a confidence interval:

$$(L, U) = \left(\bar{y} - t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})}, \bar{y} + t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} \right)$$

- Then

$$t_{1-\alpha/2, n-1} \sqrt{\text{var}(\bar{y})} = \left(\frac{U - L}{2} \right) = e$$

$$\hat{V} = \sqrt{\text{var}(\bar{y})} = \frac{U - L}{2t_{1-\alpha/2, n-1}} = \frac{e}{z_{1-\alpha/2}}$$

Alternatively...

- Desired precision: $\hat{V}^2 = \left(\frac{e}{z_{1-\alpha/2}} \right)^2 = \left(1 - \frac{n}{N} \right) \frac{S^2}{n}$
- Solve algebraically for n : $n = \frac{\hat{S}^2}{\left(\frac{e}{z_{1-\alpha/2}} \right)^2 + \frac{\hat{S}^2}{N}}$
- Ignoring the *fpc*: $n = \frac{s^2}{\left(\frac{e}{z_{1-\alpha/2}} \right)^2}$

Relative Precision Alternative

- $cv(\bar{y}) = \frac{se(\bar{y})}{\bar{y}} = \frac{\hat{V}}{\bar{y}}$
- For example, supposed that we want $cv(\bar{y}) = 0.05$ for $\bar{y} = p = 0.1$

$$\hat{V} = se(\bar{y}) = \bar{y} \times cv(\bar{y}) = 0.1 \times 0.5 = 0.005$$

- Then, calculate sample size as on previous slide:

$$n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$$

Example: adjusting a preliminary n

$$n = \frac{\hat{S}^2}{\hat{V}^2 + \frac{\hat{S}^2}{N}}$$

$$\hat{S}^2 = 6,250, \hat{V} = 2.5$$

$$N = 100,000$$

$$n = \frac{6,250}{2.5^2 + \frac{6,250}{100,000}} = 991$$

$$n' = \frac{\hat{S}^2}{\hat{V}^2}, \quad n = \frac{n'}{1 + \frac{n'}{N}}$$

$$\hat{S}^2 = 6,250, \hat{V} = 2.5$$

$$n' = \frac{\hat{S}^2}{\hat{V}^2} = \frac{6,250}{2.5^2} = 1,000$$

$$N = 100,000$$

$$n = \frac{n'}{1 + \frac{n'}{N}} = \frac{1,000}{1 + 0.01} = 991$$

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

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Frame Problems

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A. Frames

- Frame: **set of materials and/or procedures** used to designate a sample of units
- Rule links frame elements to population elements
- Accurate and up-to-date frames located in one location preferred
- Numbered, computerized lists are best.

Unit 2: Elements

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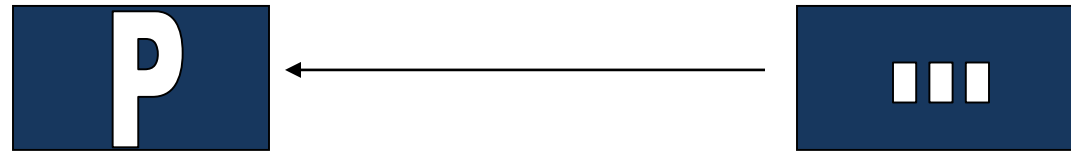
Unit 2: Elements

Frame Problems

- A. Frames
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- E. Clustering
- F. Many-to-many matching

B. Non-coverage

- Some population elements are not on the frame
 - Have zero chance of selection.



- Solution: use supplemental frames that cover non-covered elements
- Solution: use non-coverage weighting adjustments
 - Considered under post-stratification

B. Non-coverage

Target Population
Elements

Frame Population
Elements

425 Kingsley St.	←	314-555-9124
3737 Geddes Rd.		
45 S. Main, Apt. 2	←	314-555-9156
551 Seventh St.	←	314-555-8847
1845 W. Liberty	←	314-555-9173
968 Hill St.	←	314-555-9222
399 N. Fourth Ave.	←	314-555-9274
.		.
.		.
.		.

**Target Population Element Without
a Sampling Frame Element**

Unit 2: Elements

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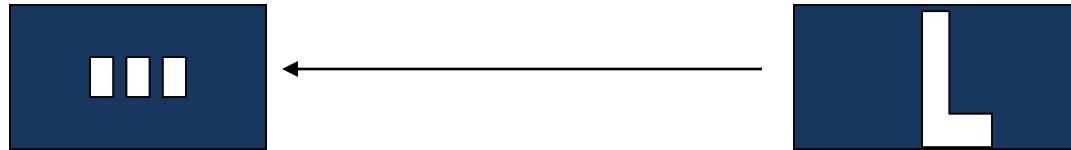
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. **Blanks**
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

C. Blanks

- Frame elements do not have corresponding population elements:



- Know frame element is blank after selection
- Screening to find eligible list elements
- One solution to **avoid**: substitute the next listing

C. Blanks

Target Population
Elements

Frame Population
Elements

425 Kingsley St.	←—————	314-555-9124
(Not in service)	←-----	314-555-1433
45 S. Main, Apt. 2	←—————	314-555-9156
551 Seventh St.	←—————	314-555-8847
1845 W. Liberty	←—————	314-555-9173
968 Hill St.	←—————	314-555-9222
399 N. Fourth Ave.	←—————	314-555-9274
.		.
.		.
.		.

**No Target Population Element for
a Sampling Frame Element**

Rejecting Foreign Elements

- Solution: reject blank
 - Leads to smaller sample size than desired
 - Adjust the sampling rate (increase number of listings):
 - Let p_b denoted estimated proportion blank elements
 - n is the desired final sample size of elements (and not list elements)
 - m is the number of frame elements to select.

$$m = \frac{n}{(1 - p_b)}$$

Unit 2: Elements

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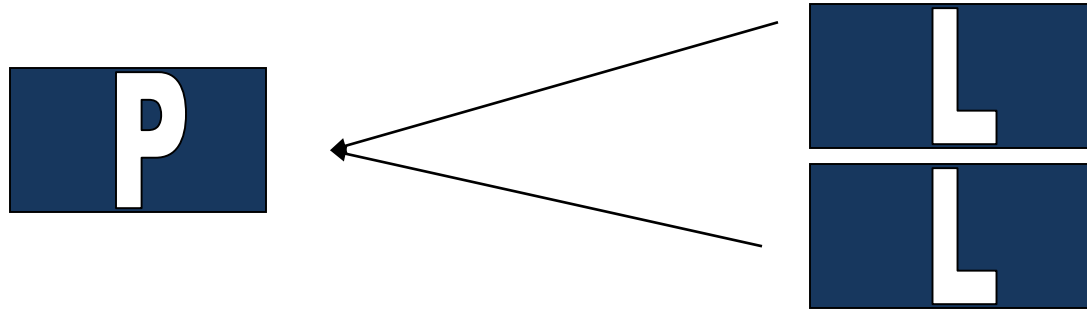
Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates**
- E. Clustering
- F. Many-to-many matching

D. Duplicates

- Occur when a single population element is linked to two or more frame elements

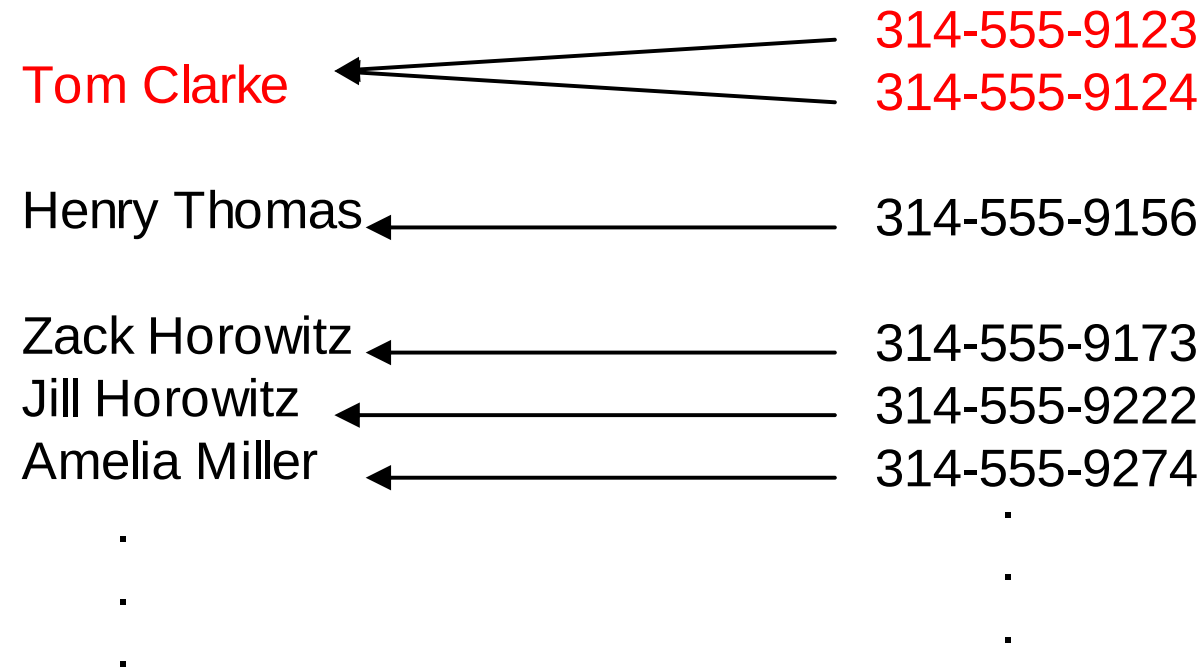


- If only a few readily identified, remove from the list before selection
- Eliminating duplicates from the sample still leaves unequal probabilities of selection

D. Duplicates

Target Population
Elements

Frame Population
Elements



**Duplication of Target Population Elements
By More Than One Sampling Frame Element**

Other solutions

- Choose unique listing
 - First, last, largest, or randomly chosen frame listing
- Determine how many duplicates for a given selected element
 - Say d_i .
- Weight observation for that selection with probability $1/d_i$.

Duplicate Weighting Illustration

- *Epsem*: $f = n/N$ for $i = 1, 2, \dots, N$
- The unweighted mean is appropriate:

$$\bar{y} = \frac{\sum_{i=1}^n y_i}{n} = \frac{y_1 + y_2 + \dots + y_n}{1+1+\dots+1}$$

- For example, from $N = 2,000$ families select $n = 25$ with *epsem*

$$f = \frac{25}{2000} = \frac{1}{80} \longrightarrow w_i = f^{-1} = 80$$

Weighted Means

- *Epsem*: $f = n/N = 1/80$ for $i = 1, 2, \dots, N$
 - The weighted mean is

$$\bar{y} = \bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} = \frac{80y_1 + 80y_2 + \dots + 80y_n}{80 + 80 + \dots + 80} = \frac{\sum_{i=1}^n y_i}{n}$$

Duplicates: Non-*epsem* Illustration

- Non-*epsem* sampling has different selection probability for elements
- For example, suppose that $n = 25$ children are selected from a list of $N = 4,000$ children enrolled in school
 - Estimating family characteristics
- For families with one child enrolled $d_i=1$

$$\pi_i = \frac{25}{4000} = \frac{1}{160} \longrightarrow w_i = \pi_i^{-1} = 160$$

Unequal Chances: Over-representation

- For families with two children enrolled $d_i=2$ and

$$\pi_i = 2 \times \frac{25}{4000} = \frac{1}{80} \longrightarrow w_i = \pi_i^{-1} = 80$$

- For families with three children enrolled $d_i=3$ and

$$\pi_i = 3 \times \frac{25}{4000} = \frac{1}{53.33} \longrightarrow w_i = \pi_i^{-1} = 53.33$$

Weighted Mean, Revisited

- Compensate for varying probabilities using the weighted estimator

$$\bar{y} = \bar{y}_w = \frac{\sum_{i=1}^n w_i y_i}{\sum_{i=1}^n w_i} = \frac{(160) y_1 + (53.33) y_2 + \dots + (80) y_n}{160 + 53.33 + \dots + 80} \neq \frac{\sum_{i=1}^n y_i}{n}$$

Unit 2: Elements

Frame Problems

- A. Frames
- B. Non-coverage
- C. Blanks
- D. Duplicates
- E. Clustering
- F. Many-to-many matching

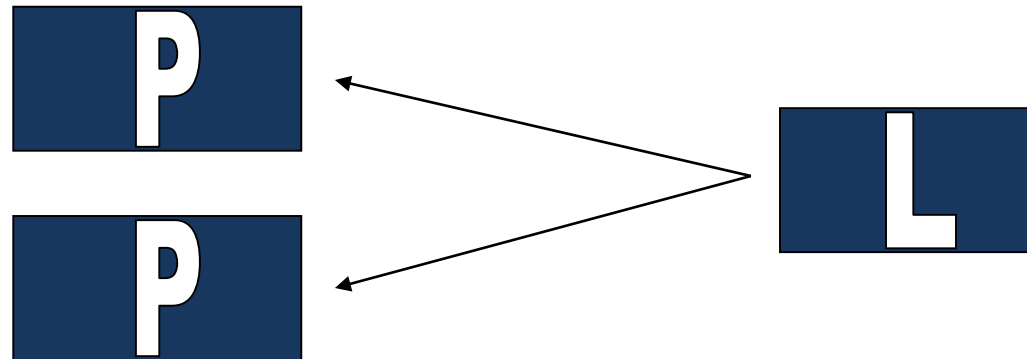
Unit 2: Elements

Frame Problems

- A. Frames
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E. Clustering

- Occurs when more than one population element can be selected by a single frame element

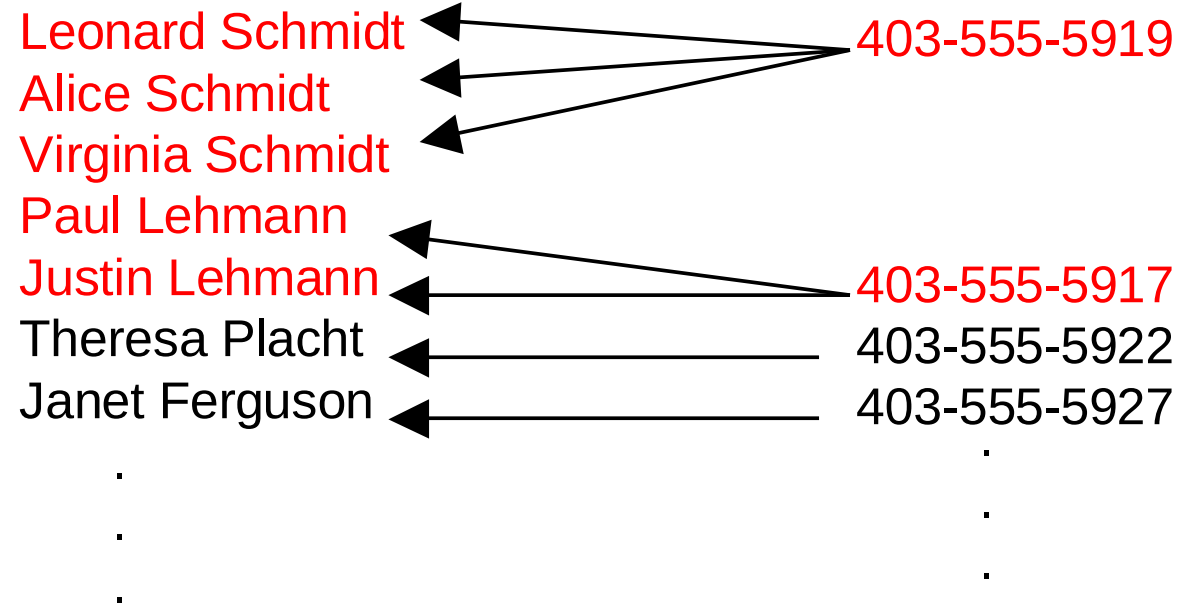


- Take all elements within selected clusters
 - Clusters unequal size: sample size varies
 - Probabilities of selection remain equal, and unweighted estimates remain unbiased

E. Clustering

Target Population
Elements

Frame Population
Elements



**Clustering Target Population
Elements Relative to Sampling Frame Elements**

Other solutions

- Take all elements in selected clusters, adjust the sample size in advance
- $m = n \times \bar{b}$
- If clustering is frequent, use cluster sampling techniques with subsampling
 - If the clusters selected with equal chances (i.e., *epsem*), subsampled elements are not
 - Weight to compensate for unequal probabilities

Objective Respondent Selection

- In the social sciences, sampling a single element from small clusters of unequal size occurs often
 - Sampling individuals from households
 - Often randomized selections are carried out by computer
- Techniques for households:
 - Objective Respondent Selection proposed by Kish
 - Rizzo method
 - Trol Dahl-Carter method
 - *(1) How many people (18 or older) live in your household... counting yourself?*
 - *(2) How many of them are men?*
 - Nearest birthday method

Unit 2: Elements

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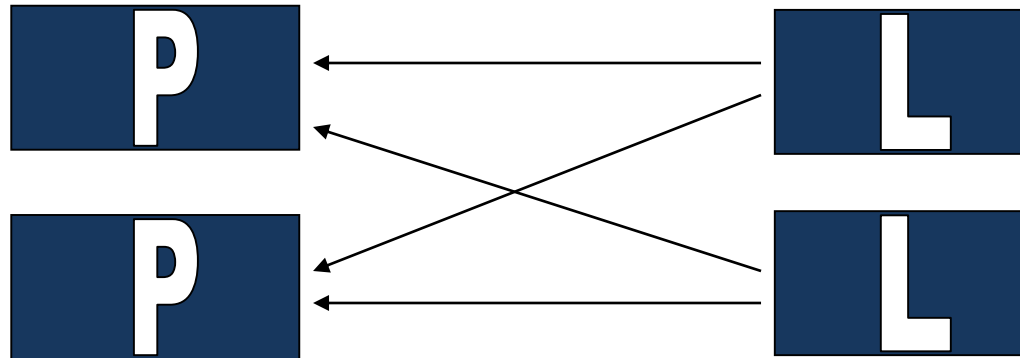
Unit 2: Elements

Frame Problems

- A. Frames
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F. Many-to-Many Matching

- Occurs when more than one population element can be selected by more than one frame element

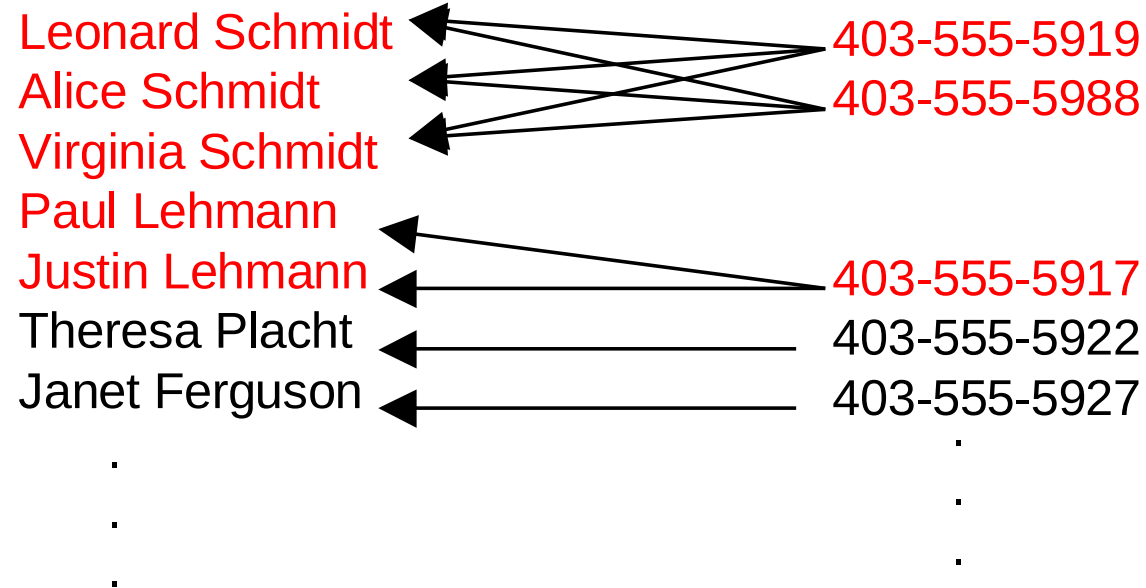


- Solution: combinations of weighting and subsampling often used

F. Many-to-Many Matching

Target Population
Elements

Frame Population
Elements



Clustering & Duplicates